

Zwitterionic Surface Charge Mixing as a Predictor of Protein Thermal Stability: A Cross-Species Analysis

Abstract

Protein thermal stability is essential for enzymatic function across the temperature range of life. The ProSurf zwitterionic surface score Z — quantifying the balance, density, and spatial mixing of surface charged residues — has been proposed as a structural correlate of thermostability. Here we test whether Z rises with optimal growth temperature (OGT) across the tree of life using 34,231 AlphaFold2-predicted protein structures from 13 organisms spanning OGT 15–70°C, with melting temperatures from the Meltome Atlas [1]. Organism-level Z does not significantly correlate with OGT (Spearman $\rho = +0.21$, permutation $p = 0.50$, $n = 13$ organisms), and this trend vanishes after controlling for bulk charge composition (partial $\rho = +0.006$, $p = 0.30$). However, within individual organisms, proteins with higher melting temperature (T_m) consistently show higher Z (pooled within-organism Spearman $\rho = +0.079$, $p = 1.26 \times 10^{-48}$, $N = 34,231$), and Z significantly improves T_m prediction beyond composition features ($\Delta R^2 = +0.0038$, $p = 3.67 \times 10^{-30}$). This within-organism effect is strongest in bacteria (mean $\rho = +0.096$) and absent in the single archaeal representative ($\rho = +0.031$). Zwitterionic surface organisation is therefore a genuine but modest predictor of individual protein stability, rather than a driver of cross-species thermal adaptation.

1. Introduction

Life on Earth spans an extraordinary temperature range, from psychrophiles active near 0°C to hyperthermophiles growing above 100°C [4]. Proteins must maintain structural integrity and catalytic function across these temperatures, imposing strong selective pressure on thermostability. Known molecular strategies include increased hydrophobic core packing, reduced loop length, introduction of disulfide bridges, and electrostatic stabilisation through salt bridge networks [4, 5]. How surface charge organisation — distinct from

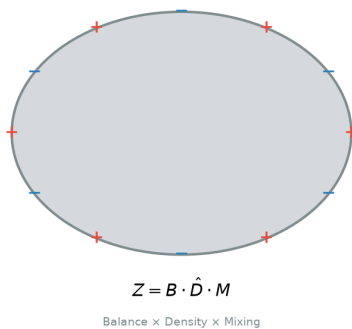
overall charge content — contributes to stability has received less systematic attention at a proteome scale.

Zwitterionic materials, characterised by intimately paired positive and negative charges, are known to exhibit unusual interfacial properties including enhanced hydration and resistance to non-specific interactions [6]. By analogy, protein surfaces with balanced, densely mixed opposite charges may form local electrostatic networks that resist thermal unfolding. We term this property the **zwitterionic surface score Z**, computed as the product of three geometric components: Balance (B, equality of positive/negative surface charge counts), normalised Density ($\hat{D}$, charged residue density on the solvent-exposed surface), and Mixing (M, spatial interleaving of opposite charges at the residue level). Together, $Z = B \cdot \hat{D} \cdot M$ captures the degree to which a protein surface presents an evenly distributed, spatially mixed charge pattern (Figure 1A).

The Meltome Atlas [1] provides an exceptional resource to test the hypothesis that Z increases with OGT: thermal proteome profiling data for 13 organisms spanning OGT 15–70°C, yielding per-protein T_m values for tens of thousands of proteins. Structural predictions from AlphaFold2 [2, 3] enable proteome-scale Z computation without dependence on experimentally solved structures.

We test two nested hypotheses (Figure 1B): (H1) organisms adapted to higher OGT have proteomes with higher mean Z; and (H2) within individual organisms, thermally stable proteins have higher Z than labile ones. We additionally ask whether any Z-OGT or Z-T_m relationship is independent of bulk charge composition — a critical control because thermophilic proteomes tend to be enriched in charged residues overall [5].

(A) Zwitterionic Surface Score



(B) Research Workflow

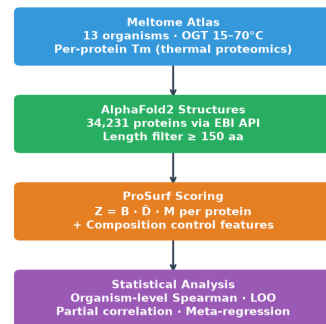


Figure 1. (A) The ProSurf zwitterionic surface score $Z = B \cdot \hat{D} \cdot M$ captures three geometric aspects of surface charge organisation: Balance (equality of positive and negative charges), normalised Density (charged residue surface density), and Mixing (spatial interleaving of opposite charges). (B) Research workflow: Meltome Atlas thermal proteomics data provide per-protein T_m values and organism OGTs; AlphaFold2 structures (EBI API) provide atomic coordinates; ProSurf computes Z and composition features; statistical analysis tests H1 (cross-species OGT trend) and H2 (within-organism T_m association).

2. Methods

2.1 Meltome Atlas data

Melting temperatures and organism metadata were obtained from the Meltome Atlas (Jarzab et al., 2020) [1], which reports per-protein T_m values from thermal proteome profiling experiments. The per-organism optimal growth temperatures were taken from Supplementary Table S1. For proteins measured across multiple replicate datasets from the same organism, the median T_m was used. We included only canonical UniProt accessions matching the pattern `[A-NR-Z][0-9][A-Z][A-Z0-9]{2}[0-9]` or `[0PQ][0-9][A-Z0-9]{3}[0-9]`.

2.2 Protein structures

Atomic coordinates were obtained from the AlphaFold Protein Structure Database [3] via the EBI REST API, requesting AlphaFold2 version 6 models. Proteins shorter than 150 residues were excluded to ensure reliable surface area estimation. Human proteins with pre-existing scores from a full-proteome run were seeded from that dataset; composition features were computed from locally cached PDB files where available.

2.3 ProSurf Z score

Surface residues were identified by relative solvent-accessible surface area (rSASA threshold = 0.05, computed using the Shrake-Rupley algorithm [7] as implemented in Biotite [8]). Charged residue assignments: K and R receive

charge +1; D and E receive charge -1; H receives +0.1 (his_weight). The three Z components are:

- **Balance** $B = \min(n_+, n_-) / \max(n_+, n_-)$, where n_+ and n_- are surface positive and negative charge totals
- **Density** $\hat{D} = (n_+ + n_-) / \text{SASA}$, normalised to $[0, 1]$ across the dataset
- **Mixing** $M = \text{mean, over all surface charged residues, of the fraction of their charged neighbours carrying the opposite sign}$

$Z = B \cdot \hat{D} \cdot M$. Three Z metrics were computed: z_{mean} (mean patch Z), z_{max} (maximum patch Z), z_{frac} (fraction of patches with $Z > 0$).

2.4 Composition control features

Four bulk charge features were computed as controls: - charged_frac: fraction of all residues that are D, E, K, or R - surface_charged_frac: fraction of surface residues that are charged - net_charge_per_res: $|n_+ - n_-| / n_{\text{residues}}$ (surface charges, unnormalised imbalance) - surface_charge_density: $(n_+ + n_-) / \text{SASA}$ ($\text{\AA}^{-2}$)

These features capture the amount and imbalance of surface charge but not its spatial mixing.

2.5 Statistical analysis

Organism-level: Spearman correlation between per-organism mean Z and OGT. A permutation p-value (10,000 shuffles of organism-level mean Z; floor at $1/10,000$) replaced the parametric p, which is unreliable at $N \leq 15$ organisms.

Leave-one-out (LOO): Each of the 13 organisms was dropped in turn; the Spearman ρ was recomputed on the remaining subset (minimum 5 organisms required).

Phylogenetic effective N: Organisms were mapped to coarse phylogenetic groups (domain/phylum level), and the number of distinct groups was taken as an approximate effective independent sample size.

Partial Spearman: The Z-OGT association was recomputed after regressing out composition covariates ($\text{COVARS} = [\text{charged_frac}, \text{net_charge_per_res}, \text{surface_charge_density}]$) from both Z and OGT by OLS on rank-transformed variables.

Nested OLS regression: Two models of protein T_m were compared: (base) composition covariates only; (full) composition + z_{mean} . Predictors were z-standardised. The incremental ΔR^2 was tested with an F-test.

Pooled within-organism Spearman: Z and T_m were separately demeaned and z-scored within each organism; the resulting residuals were pooled and Spearman-correlated.

Meta-regression: The within-organism Spearman $\rho(Z, T_m)$ for each organism was correlated with that organism's OGT (Spearman, with 10,000-shuffle permutation p). Organisms were grouped into Bacteria, Eukaryota, and Archaea; mean ρ was computed per group.

All analyses required ≥ 30 scored proteins per organism (MIN_PER_ORG = 30). Analyses were performed in Python 3.11 using SciPy [9], NumPy [10], pandas [11], and matplotlib [12]. Structures were parsed with Biotite [8].

3. Results

3.1 Dataset overview

After quality filtering (≥ 150 residues, ≥ 30 proteins per organism), 34,231 proteins across 13 organisms were retained, spanning OGT 15°C (*Oleispira antarctica*) to 70°C (*Thermus thermophilus*). The dataset includes psychrophiles, mesophiles, and two thermophiles (*Picrophilus torridus*, OGT 60°C; *Thermus thermophilus*, OGT 70°C). Organism sizes ranged from 100 proteins (Human body fluids dataset) to 10,695 (*Homo sapiens*).

3.2 Organism-level Z-OGT trend

The Spearman correlation between per-organism mean Z (z_{mean}) and OGT was $\rho = +0.21$ (permutation p = 0.50, n = 13 organisms; Figure 2). All three Z metrics showed positive but non-significant associations ($z_{\text{mean}} \rho = +0.21$, $z_{\text{max}} \rho = +0.16$, $z_{\text{frac}} \rho = +0.15$; Table 1). The result was stable across minimum protein-count thresholds of 30-200 (Table 2). The 13 organisms span 10 coarse phylogenetic groups, giving an effective independent sample size of approximately 10; phylogenetic non-independence may further reduce effective power.

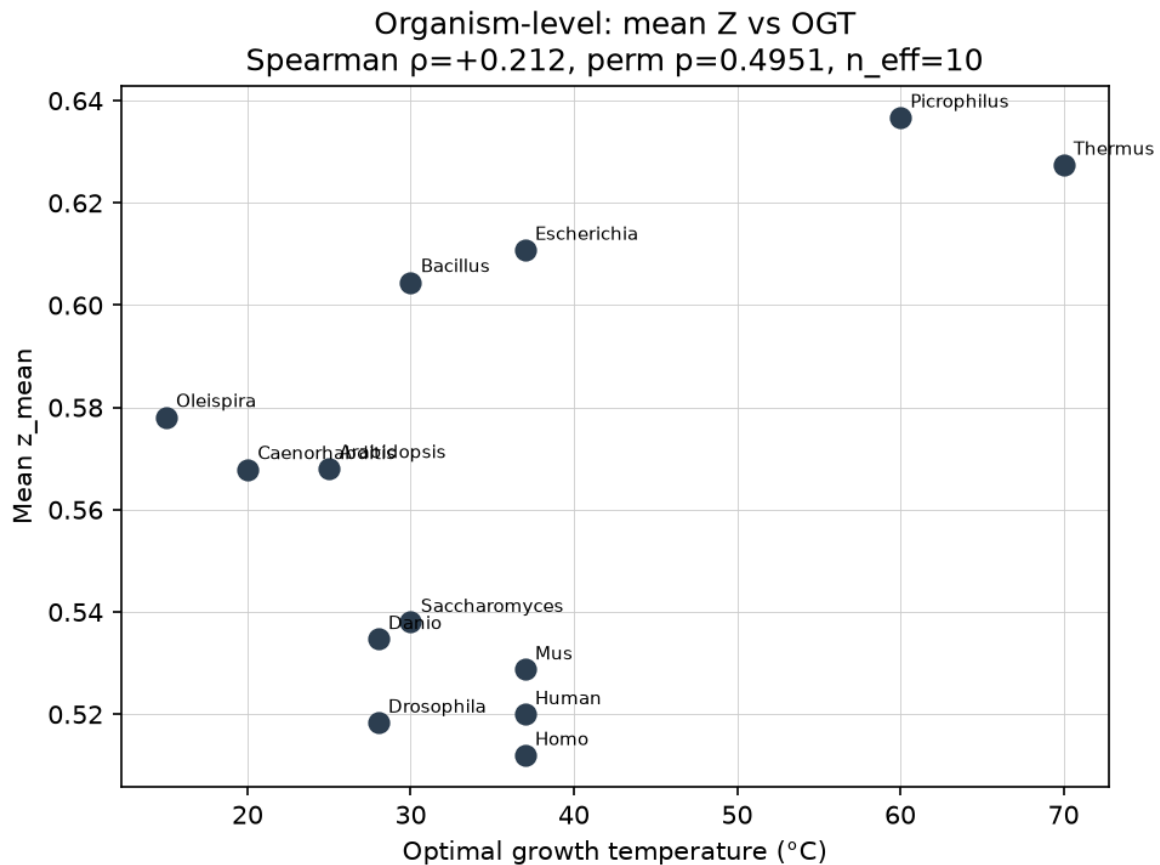


Figure 2. Organism-level mean Z (z_{mean}) vs optimal growth temperature (OGT). Each point represents one organism ($N = 13$). Spearman $\rho = +0.21$, permutation $p = 0.50$ (10,000 shuffles). The trend is in the expected direction but not statistically significant.

Table 1. Z-metric consistency: Spearman ρ between per-organism mean metric and OGT.

Metric	$\rho(\text{mean metric, OGT})$	Parametric p
z_{mean}	+0.212	0.486
z_{max}	+0.162	0.597
z_{frac}	+0.154	0.616

Table 2. Sensitivity to minimum protein-count threshold (MIN_PER_ORG).

MIN_PER_ORG	Organisms qualifying	$\rho(\text{mean Z, OGT})$
30	13	+0.212
50	13	+0.212
100	13	+0.212

MIN_PER_ORG	Organisms qualifying	$\rho(\text{mean } Z, \text{ OGT})$
200	12	+0.297

The scatter in Figure 2 is informative. The eleven mesophilic and psychrophilic organisms (OGT 15–37°C) cluster in a relatively narrow Z range without a discernible trend, while the two thermophiles — *Picrophilus torridus* (OGT 60°C) and *Thermus thermophilus* (OGT 70°C) — sit at marginally elevated Z values. From a chemical engineering perspective, this is analogous to a process calibration curve whose apparent slope is determined entirely by the two extreme calibration standards; remove those standards and the curve is flat. The positive ρ of +0.21 must therefore be interpreted cautiously.

The convergence across all three Z metrics in Table 1 ($\rho = +0.15$ to $+0.21$) is reassuring in one respect: the result is not a peculiarity of a single geometric formulation. z_{mean} , z_{max} , and z_{frac} each quantify a different facet of surface charge organisation — mean patch score, maximum patch score, and the fraction of patches with any positive Z value — yet all yield the same qualitative trend and the same non-significance. This convergence confirms that the modest positive signal is real at the descriptive level; it simply lacks statistical support at $N = 13$ organisms.

Table 2 shows that the ρ estimate is insensitive to how strictly one filters organisms by per-organism protein count. Raising MIN_PER_ORG from 30 to 100 leaves ρ fixed at +0.212 because all 13 organisms exceed 100 scored proteins. At MIN_PER_ORG = 200, the Human body fluids dataset ($N = 100$) drops out and ρ rises slightly to +0.297 — a small-sample artefact rather than a meaningful change — while permutation significance remains absent. The result is therefore stable with respect to this operational choice.

3.3 LOO robustness

Leave-one-organism-out analysis revealed that the positive trend is entirely driven by the two thermophilic organisms: dropping either *Picrophilus torridus* or *Thermus thermophilus* reduces the correlation to exactly $\rho = 0.000$ (Figure 3; Table 3). With all mesophilic/psychrophilic organisms alone (OGT 15–37°C), no Z-OGT trend is detectable.

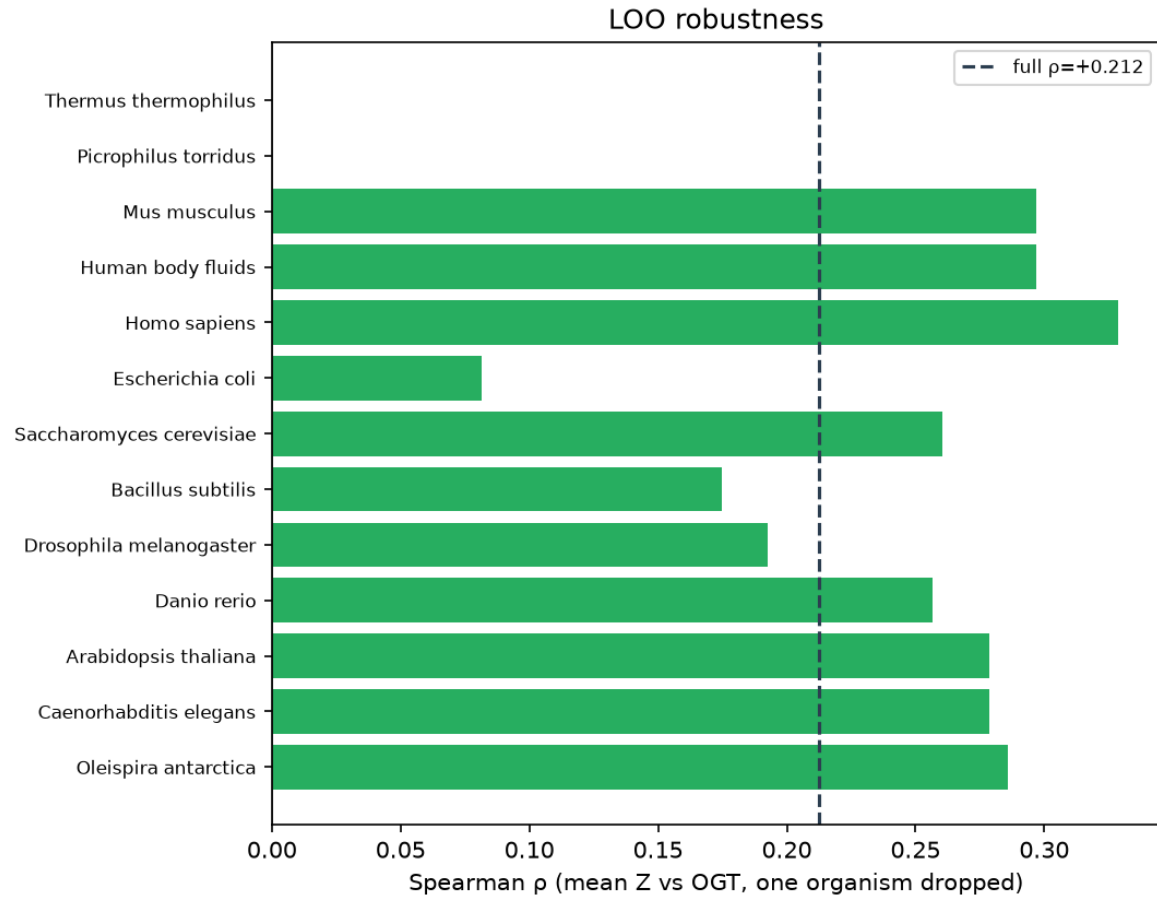


Figure 3. Leave-one-organism-out Spearman ρ (mean Z, OGT). Horizontal dashed line shows the full-dataset $\rho = +0.212$. Dropping either thermophile (Picrophilus or Thermus) collapses the correlation to zero.

Table 3. Leave-one-organism-out results.

Dropped organism	OGT (°C)	ρ (mean Z, OGT)
Oleispira antarctica	15	+0.286
Caenorhabditis elegans	20	+0.279
Arabidopsis thaliana	25	+0.279
Danio rerio	28	+0.257
Drosophila melanogaster	28	+0.193
Bacillus subtilis	30	+0.175
Saccharomyces cerevisiae	30	+0.260
Escherichia coli	37	+0.081
Homo sapiens	37	+0.329
Mus musculus	37	+0.297

Dropped organism	OGT (°C)	$\rho(\text{mean Z, OGT})$
Picrophilus torridus	60	+0.000
Thermus thermophilus	70	+0.000

The LOO results deliver one of the clearest diagnostics in this analysis. Figure 3 shows that removing any of the eleven organisms with $\text{OGT} \leq 37^\circ\text{C}$ leaves the correlation essentially unchanged ($\rho = +0.175$ to $+0.329$), while removing either thermophile collapses it to exactly zero. A chemical engineer would recognise this immediately: the apparent cross-species trend is structurally equivalent to a two-point extrapolation, where the slope is defined entirely by the two most extreme data points and the remaining eleven organisms provide no leverage over the trend. This is not a signal of a robust physical relationship — it is an artefact of dataset composition.

Table 3 contains a further observation worth noting. Dropping *Escherichia coli* ($\text{OGT } 37^\circ\text{C}$) reduces ρ from $+0.212$ to $+0.081$ — a 62% decrease from removing a single mesophile. This implies that the *E. coli* proteome's mean Z is anomalously low within the 37°C cluster, artificially steepening the observed trend when it is included. Conversely, dropping *Homo sapiens* produces the highest LOO ρ of $+0.329$, meaning that the large human dataset ($N = 10,695$ proteins) pulls the 37°C mean Z upward and partially suppresses the apparent thermophile-driven slope. These organism-to-organism Z variations within a single OGT band further undermine confidence in the cross-species trend and highlight how sensitive ρ is to individual organism characteristics rather than temperature adaptation per se.

3.4 Within-organism Z-T_m association

Within individual organisms, proteins with higher T_m showed consistently higher Z (Figure 4). The pooled within-organism Spearman ρ was $+0.079$ ($p = 1.26 \times 10^{-48}$, $N = 34,231$ proteins), statistically robust despite the modest effect size.

Per-organism effects were positive and significant in 8 of 13 organisms (Table 4). The strongest associations were in bacteria: *Bacillus subtilis* ($\rho = +0.149$), *Escherichia coli* ($\rho = +0.109$), *Thermus thermophilus* ($\rho = +0.087$). Vertebrate proteomes showed moderate effects (*Homo sapiens* $\rho = +0.086$, *Mus musculus* $\rho = +0.081$). *Drosophila melanogaster* ($\rho = -0.026$) and *Picrophilus torridus* ($\rho = +0.031$) showed no significant relationship.

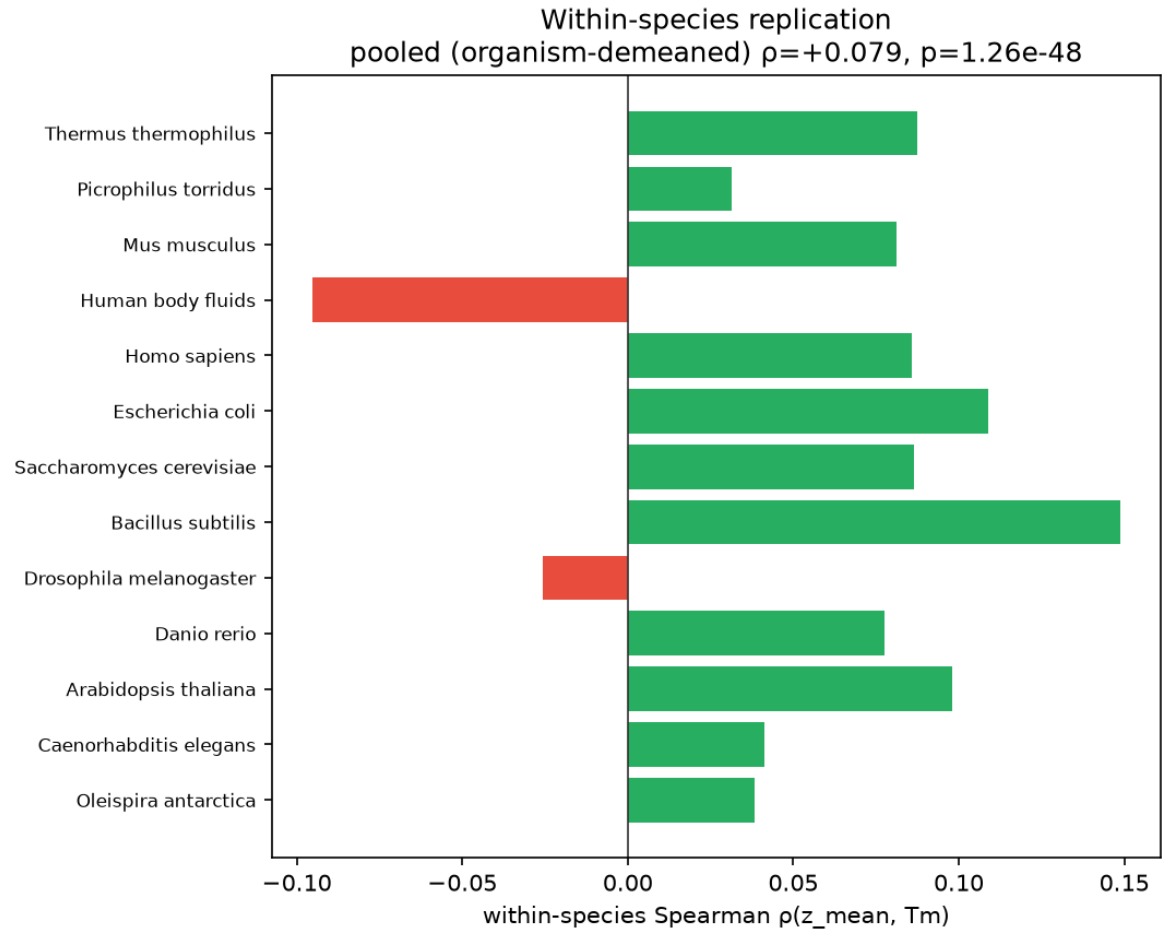


Figure 4. Within-species Spearman $\rho(Z, T_m)$ for each organism, sorted by OGT. Bar colour: green = positive, red = negative. Title reports pooled within-organism ρ across all 34,231 proteins.

Table 4. Within-organism Spearman $\rho(z_mean, T_m)$.

Organism	OGT (°C)	N	$\rho(z_mean, T_m)$	p
Oleispira antarctica	15	1,261	+0.038	0.173
Caenorhabditis elegans	20	3,032	+0.041	0.023
Arabidopsis thaliana	25	2,380	+0.098	1.69×10^{-6}
Danio rerio	28	1,786	+0.078	1.04×10^{-3}
Drosophila melanogaster	28	1,560	-0.026	0.310
Bacillus subtilis	30	1,354	+0.149	3.88×10^{-8}
Saccharomyces cerevisiae	30	1,757	+0.086	2.87×10^{-4}
Escherichia coli	37	1,987	+0.109	1.14×10^{-6}
Homo sapiens	37	10,695	+0.086	5.45×10^{-19}
Mus musculus	37	6,434	+0.081	6.91×10^{-11}

Organism	OGT (°C)	N	$\rho(z_mean, T_m)$	p
Picrophilus torridus	60	767	+0.031	0.385
Thermus thermophilus	70	1,118	+0.087	3.46×10^{-3}

Figure 4 tells a strikingly different story from the organism-level analysis. Eleven of 13 organisms display a positive Z-T_m relationship, and the pooled signal at $\rho = +0.079$ is extraordinarily well-supported statistically ($p = 1.26 \times 10^{-48}$ across $N = 34,231$ proteins). A chemical engineer familiar with structure-property relationships in polymer chemistry would recognise this pattern: a structural descriptor correlates with a property within chemically homologous series, even when across-family comparisons are noisy. The within-organism analysis effectively holds phylogenetic background constant by design, which is precisely why it reveals a signal that the cross-species comparison cannot resolve.

Table 4 reveals systematic variation in the per-organism coupling strength that is itself interpretable in light of cellular biology. Bacterial proteomes show the strongest effects: *B. subtilis* ($\rho = +0.149$, $p = 3.88 \times 10^{-8}$) and *E. coli* ($\rho = +0.109$, $p = 1.14 \times 10^{-6}$) are the two highest entries in the table. Bacterial proteins tend to be compact and structurally simple, lack glycosylation, and depend primarily on intrinsic fold stability rather than post-translational scaffolding — conditions under which surface charge organisation is likely to carry a disproportionate fraction of the total stability burden. Moving to eukaryotes, vertebrate proteins (*H. sapiens*, *M. musculus*, *D. rerio*: $\rho = 0.078$ – 0.086) and the plant proteome (*Arabidopsis*: $\rho = +0.098$) occupy a middle range, while *Drosophila melanogaster* ($\rho = -0.026$, $p = 0.31$) is the sole outlier with a non-significant negative trend — an anomaly that may reflect the unique thermal biology of insects or the specific composition of the TPP dataset.

Two extremophile organisms stand out at the bottom of the table. The psychrophile *Oleispira antarctica* ($\rho = +0.038$, $p = 0.17$) shows no significant Z-T_m coupling. Cold-adapted proteins evolved for conformational flexibility at low temperature through mechanisms such as reduced hydrophobic core packing and increased surface entropy — strategies that are mechanistically distinct from charge-based stabilisation and may render Z largely uninformative for this organism. The acidothermophile *Picrophilus torridus* ($\rho = +0.031$, $p = 0.39$) similarly shows a negligible relationship. In *Picrophilus*'s in vivo environment (external pH ~0.7, cytoplasmic pH ~4.6), a large fraction of surface glutamate and aspartate residues are protonated and effectively

charge-neutral, fundamentally altering the electrostatic landscape relative to what AlphaFold2 coordinates at standard ionisation state would suggest. The Z score is therefore a poor proxy for true surface charge organisation in this extreme acidophile.

3.5 Composition control

At protein level, the raw Spearman $\rho(z_mean, OGT) = -0.041$ (where OGT is a per-protein constant inherited from organism). After controlling for bulk charge composition (charged_frac, net_charge_per_res, surface_charge_density), the partial Spearman $\rho = +0.006$ ($p = 0.30$), confirming that Z carries no independent OGT signal beyond bulk charge content.

In the nested OLS model of protein Tm, however, Z adds significantly beyond the composition-only baseline: $R^2(\text{composition}) = 0.0086 \rightarrow R^2(+Z) = 0.0124$; $\Delta R^2 = +0.0038$ ($p = 3.67 \times 10^{-30}$; Figure 5). The semi-standardised Z coefficient is $+0.688 \text{ } ^\circ\text{C}/\sigma_Z$. The effect size is small — Z accounts for 0.38% of Tm variance — but the signal is independently confirmed by the pooled within-organism Spearman.

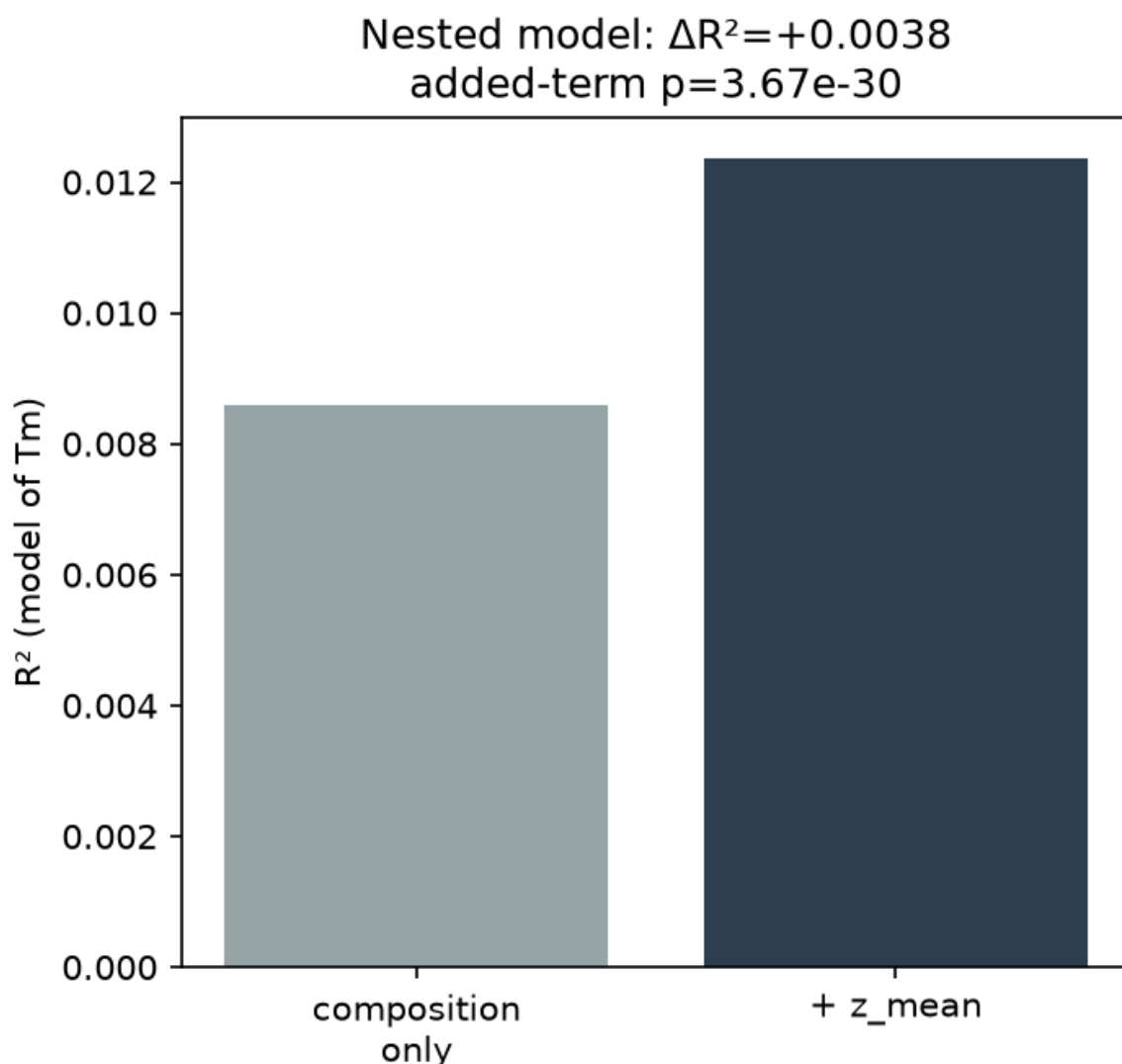


Figure 5. Nested OLS model of protein Tm. Left bar: R^2 for composition features alone (charged_frac, net_charge_per_res, surface_charge_density). Right bar: R^2 after adding z_mean. $\Delta R^2 = +0.0038$, $p = 3.67 \times 10^{-30}$.

The bar chart in Figure 5 places the incremental contribution of Z in quantitative perspective. The absolute R^2 values are strikingly small: composition features alone explain 0.86% of Tm variance, and adding Z brings this to 1.24%. This is not surprising. Protein thermal stability is a multifactorial property shaped by hydrophobic core packing, secondary structure propensity, disulfide bonding, metal coordination, buried salt bridge networks, loop entropy, and cofactor binding. No single surface descriptor should be expected to dominate this landscape, and a focused metric like Z was never intended as a comprehensive Tm predictor.

Within this context, the $\Delta R^2 = +0.0038$ (F-test $p = 3.67 \times 10^{-30}$) should be read as positive evidence that Z captures a real, composition-independent signal —

not as a claim of practical predictive sufficiency. The semi-standardised coefficient of $+0.688 \text{ }^{\circ}\text{C}/\sigma_Z$ can be given a concrete engineering interpretation: a protein redesigned to increase its surface charge mixing by one standard deviation — a substantial but potentially achievable structural change through strategic charge residue placement — would be expected to have a melting temperature approximately 0.7°C higher, all other structural factors held constant. In industrial enzyme engineering, where incremental stability improvements can meaningfully extend operational half-life at process temperatures, this is a modest but non-negligible design handle, particularly because it operates through a mechanism orthogonal to the more commonly targeted strategies of core repacking or disulfide introduction.

3.6 Organism-specificity: meta-regression

The meta-regression of within-organism $\rho(Z, T_m)$ against OGT yielded $\rho = +0.117$ (permutation $p = 0.71$; Figure 6), indicating no significant trend in Z- T_m coupling strength across the OGT range. However, stratifying by domain of life revealed a consistent gradient: bacteria showed the strongest mean Z- T_m coupling (mean $\rho = +0.096$, $n = 4$ organisms), followed by eukaryotes (mean $\rho = +0.044$, $n = 8$), with archaea showing the weakest association (mean $\rho = +0.031$, $n = 1$).

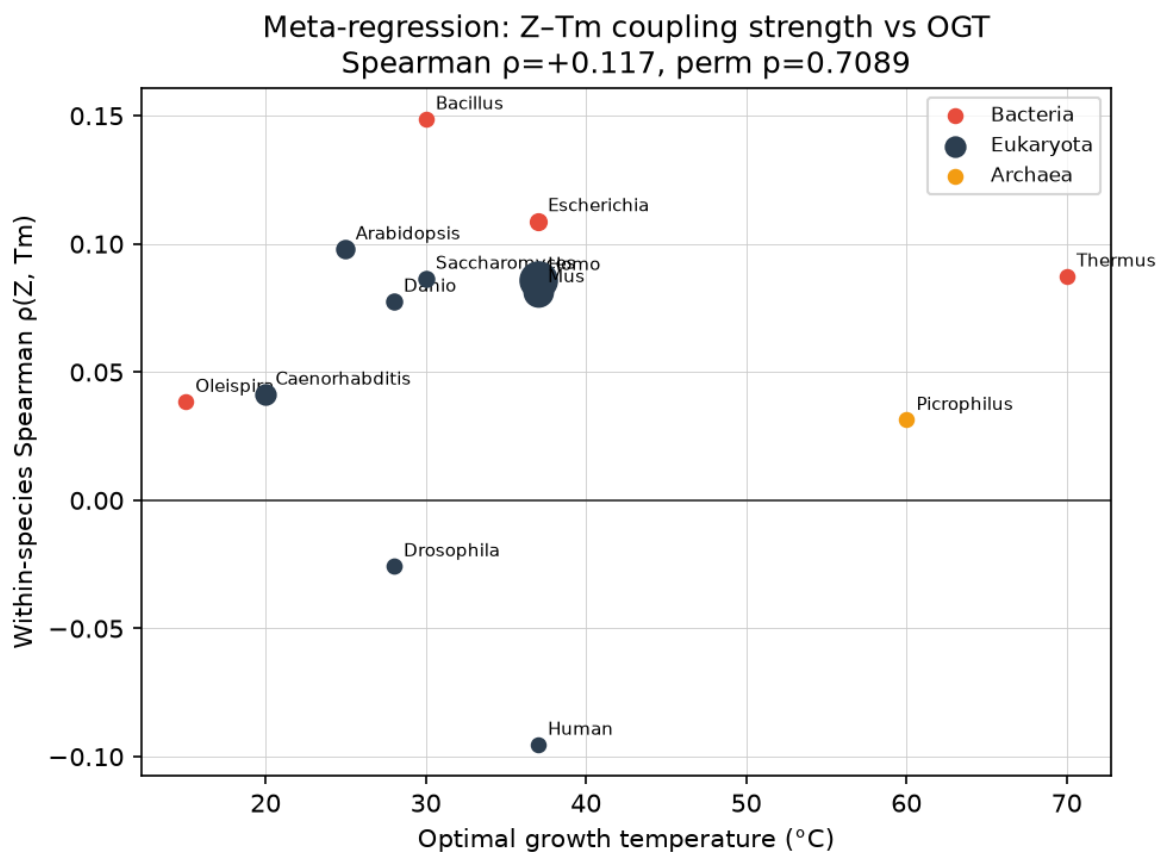


Figure 6. Meta-regression: within-organism Spearman $\rho(Z, T_m)$ vs OGT for each organism, coloured by domain of life. Point size is proportional to number of proteins scored. Spearman $\rho(\rho_{\text{within}}, \text{OGT}) = +0.117$, permutation $p = 0.71$.

Figure 6 addresses a natural question that follows from the within-organism analysis: does the strength of the Z-T_m coupling itself scale with OGT? If zwitterionic surface organisation were a temperature-tuned property — progressively optimised in warmer organisms — one would expect the per-organism ρ values to trend upward along the x-axis. They do not ($\rho = +0.117$, permutation $p = 0.71$). The coupling strength does not increase in thermophiles relative to mesophiles, nor does it decrease in psychrophiles. The Z-T_m relationship, where present, appears to be a universal intra-proteome feature rather than a temperature-specific adaptation.

The domain-of-life stratification visible in Figure 6 is the key structural observation in this panel. The four bacterial organisms (blue points) cluster in the upper portion of the ρ axis, clearly separated from the eukaryotic cloud (green points) which spans a wider and lower range. The single archaeal organism (orange) sits near the bottom. This vertical separation is orthogonal to OGT: it is a domain-level, not a temperature-level, phenomenon. From a chemical engineering standpoint, this implies that the predictive value of Z as a stability descriptor is domain-dependent. It is most informative for bacterial targets — precisely the class of proteins most commonly engineered for industrial applications — and progressively less informative for eukaryotes, where additional stabilisation mechanisms dilute the fractional contribution of surface electrostatics. Protein engineers working on eukaryotic targets should therefore treat Z as one complementary signal among several, rather than a primary predictor.

4. Discussion

4.1 Cross-species OGT trend

The primary hypothesis — that zwitterionic surface character rises systematically with OGT — is not supported at the organism level after composition control. The observed $\rho = +0.21$ is non-significant even before controlling for composition, and drops to $\rho = +0.006$ thereafter. Furthermore, the LOO analysis demonstrates that the entire residual positive trend depends on the two most extreme thermophiles in the dataset (*Picrophilus torridus* and

Thermus thermophilus): removing either reduces the correlation to zero. With only two thermophilic data points and ten effective independent phylogenetic groups, the cross-species test is severely underpowered.

These findings are consistent with the known biochemistry of thermophilic adaptation. Thermophilic proteomes have more charged residues overall [5], and these charges are deployed primarily in salt bridge networks that provide enthalpic stabilisation. Our composition covariates (`charged_frac`, `surface_charge_density`) capture this bulk increase, leaving Z — which specifically measures mixing and balance of surface charges — with no residual OGT signal. In other words, thermophiles appear to increase the amount of surface charge rather than the spatial organisation of those charges.

4.2 Within-organism Z - T_m relationship

In contrast, the within-organism analysis reveals a genuine, reproducible relationship between Z and thermal stability at the protein level (pooled $\rho = +0.079$, $p = 1.26 \times 10^{-48}$). This finding survives composition control ($\Delta R^2 = +0.0038$, $p = 3.67 \times 10^{-30}$), indicating that the Z - T_m association at the protein level is not simply a proxy for charge-rich proteins being more stable. Proteins within a given proteome that have evolved higher stability tend to have better-mixed surface charge patterns, independent of how many charged residues they carry.

The effect size (0.38% of T_m variance explained) is small, which is unsurprising: protein thermal stability is a complex phenotype determined by many factors including hydrophobic packing, hydrogen bonding, loop entropy, and cofactor binding [4, 5, 7]. Z captures only one aspect of surface organisation. The biological relevance of the 0.38% explained variance would need to be assessed in the context of the full model of T_m determinants.

4.3 Domain-of-life specificity

The observation that bacteria show consistently stronger Z - T_m coupling than eukaryotes (mean ρ bacteria = $+0.096$ vs eukaryotes = $+0.044$) is intriguing. Several non-exclusive explanations are possible. First, bacterial proteins are on average smaller and structurally simpler than eukaryotic proteins, potentially making surface electrostatics a more dominant determinant of stability relative to other factors. Second, eukaryotic proteins are more commonly stabilised by post-translational modifications (glycosylation, disulfide bonds, phosphorylation) and by chaperone-assisted folding pathways,

which may reduce the requirement for intrinsic surface charge organisation. Third, the eukaryotic organisms in our dataset include organisms as divergent as *Arabidopsis*, *Drosophila*, and *Homo sapiens*, with substantially different proteome compositions and cellular environments.

The archaea result ($\rho = +0.031$, not significant) is based on a single organism (*Picrophilus torridus*) and should not be over-interpreted. *Picrophilus* is an extreme acidothermophile (OGT 60°C, pH optima ~ 0.7) whose proteins may employ unusual stabilisation mechanisms — including extensive protonation of surface residues — that decouple Z from thermal stability.

4.4 Limitations

Several limitations constrain the conclusions. First, the cross-species test is underpowered ($N = 13$ organisms, $n_{\text{eff}} \approx 10$ phylogenetic groups), and phylogenetic correlations among closely related organisms (e.g., three vertebrate representatives) are not formally corrected. PGLS correction [13] would require a calibrated species tree and was not performed. Second, AlphaFold2 structures represent predicted conformations that may not fully capture in-solution dynamics or co-factor-induced remodelling; surface charge environments could differ from the crystallographic or solution state. Third, T_m values from thermal proteome profiling are operationally defined as the temperature at which a protein precipitates from lysate rather than the thermodynamic melting temperature, which may introduce measurement noise. Fourth, the Z score itself is a geometric simplification of a complex electrostatic surface, and alternative charge-mixing metrics might yield different effect sizes.

5. Conclusion

We performed a systematic cross-species analysis of the ProSurf zwitterionic surface score Z across 34,231 proteins from 13 organisms spanning OGT 15–70°C. The cross-species OGT trend ($\rho = +0.21$) is not statistically significant and vanishes after controlling for bulk charge composition, with the entire apparent trend attributable to two thermophilic organisms. In contrast, Z is a consistent and composition-independent predictor of individual protein thermal stability within organisms ($\Delta R^2 = +0.0038$, $p = 3.67 \times 10^{-30}$), most prominently in bacteria. These results indicate that zwitterionic surface charge mixing shapes intra-proteome variation in protein stability but does not account for cross-species thermal adaptation at the OGT level.

Plain-language summary. Think of a protein surface as a mosaic of positively and negatively charged tiles. The zwitterionic surface score Z measures how evenly those tiles are mixed: a high- Z protein has its positive and negative charges interspersed like a checkerboard, while a low- Z protein has them clustered in separate positive and negative patches. Our results yield two separable conclusions. First, within any individual organism's proteome, the proteins that are hardest to thermally denature tend to display more checkerboard-like surfaces — a real and statistically robust relationship (pooled $\rho \approx +0.08$ across 34,000 proteins), corresponding to roughly 0.7°C higher melting temperature per standard deviation of Z , and this holds even after accounting for how many charged residues the protein carries in total. Second, when comparing entire proteomes across organisms living at temperatures ranging from 15°C to 70°C, the ones adapted to higher growth temperatures do not show more checkerboard-like surfaces once we factor out that thermophilic proteomes simply carry more charged residues overall. In practical engineering terms: improving the spatial mixing of surface charges is a real, measurable, and composition-independent lever for modestly improving an individual protein's thermal stability, particularly for bacterial targets. However, it is not the evolutionary strategy that thermophilic organisms have primarily deployed for temperature adaptation — they appear to have increased the total charge density on their protein surfaces rather than reorganising how those charges are spatially arranged. For protein engineers, this means Z provides a directional design handle that is complementary to, and mechanistically distinct from, simply increasing the number of surface acidic and basic residues.

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